

Notes: Fouka (RES, 2020)

Backlash: The Unintended Effects of Language Prohibition in U.S. Schools after World War I

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1 Big picture

- **Question.** Do forced assimilation policies make minority groups more integrated, or can they provoke backlash that strengthens minority identity?
- **Policy setting.** After World War I, several U.S. states restricted or prohibited German-language instruction in schools. The paper focuses on Indiana and Ohio, which banned German in schools in 1919, and compares them to neighbouring border counties in Illinois, Michigan, Kentucky, West Virginia, and Pennsylvania.
- **Treatment group.** The main treated population is second-generation German-American males born 1880–1916, living in border counties in 1920. Treated cohorts are those young enough to have been subject to compulsory schooling when the ban was in force.
- **Research design.** The empirical design is a difference-in-differences comparison of exposed and older cohorts across law and no-law states, combined with a border-county sample that reduces geographic heterogeneity.
- **Outcomes.** The paper measures later-life ethnic identity using three behaviour-based outcomes: the German distinctiveness of children’s names, whether the individual married a German spouse, and whether the individual volunteered for the U.S. Army in World War II.
- **Main finding.** The policy did not assimilate German-Americans. It produced backlash: affected German-American cohorts gave children more German names, were more likely to marry within the German ethnic group, and were less likely to volunteer in WWII.
- **Magnitude.** Treated cohorts are about 3.6–5.7 percentage points more likely to marry endogamously and about 2.5 percentage points less likely to volunteer. Naming effects are large enough to move from a neutral name to a distinctly German name.
- **Mechanism.** The backlash is consistent with parental cultural transmission. If schools stop transmitting minority culture, parents with strong minority identity may invest more at home or through ethnic institutions.
- **Heterogeneity.** The backlash is weaker for children of mixed couples, stronger in counties with more Lutherans, and stronger where Germans are a smaller minority share. These patterns match a model in which parental and peer socialization are substitutes.
- **Church evidence.** Lutheran Church–Missouri Synod Sunday-school enrolment rises more in law states after the ban. Other church activities do not show the same pattern, suggesting increased parental demand for enculturation rather than a general church supply shock.
- **Education and earnings.** The language ban slightly increases schooling but weakly reduces later wage income. This supports the view that the backlash is cultural, not simply a by-product of reduced education.
- **Takeaway.** Assimilation policies can backfire when they threaten identity. Forced monolingualism can turn a school-integration policy into a signal that motivates parents and communities to intensify ethnic transmission.

2 Historical background

2.1 German-Americans and German language schooling

- Germans were the largest immigrant-origin group in the United States in the late nineteenth and early twentieth centuries.
- Before World War I, German language instruction was common in many states. German was taught in public schools, private schools, and parochial schools.

- German-speaking communities were especially visible in the Midwest and in Lutheran and Catholic institutional networks.
- German schooling served two functions: ordinary education and cultural transmission. It helped children learn while also preserving family language, ethnic identity, and religious community ties.
- This means that a ban on German was not just a curriculum reform. It changed one of the institutional channels through which families transmitted ethnic identity.

Interpretation: The policy shock matters because German schooling was part of the socialization technology of German-American families. Removing German from schools could either assimilate children into English-speaking culture or trigger compensating parental investment in German identity.

2.2 World War I, anti-Germanism, and language restrictions

- When the United States entered World War I, anti-German sentiment intensified sharply.
- German language, German cultural organizations, and German names became politically suspicious.
- Language restrictions were justified as patriotic measures that would Americanize immigrant children and make them more loyal citizens.
- Indiana and Ohio enacted German bans in schools in 1919. The paper exploits the fact that these treated states border several similar states without comparable German bans.

Interpretation: The timing is plausibly driven by wartime nationalism rather than gradual local assimilation trends. This makes the episode useful for studying the consequences of forced assimilation, but the paper still must address discrimination and confounding anti-German laws.

2.3 Why the border design matters

- Indiana and Ohio border Illinois, Michigan, Kentucky, West Virginia, and Pennsylvania.
- The paper restricts attention to counties along these borders to improve comparability.
- Border counties share regional labour markets, migration patterns, agricultural conditions, and Midwestern settlement histories.
- Table 1 shows balance across law and no-law border counties in population density, urban share, foreign-born share, German-born share, and Lutheran share.

Interpretation: The border design does not make treatment random, but it reduces the chance that estimated effects are driven by broad differences between treated and untreated regions.

3 Data and measurement

3.1 Language-law data

- The policy variable records whether an individual’s state had a German language law and whether the individual was in the compulsory-schooling age range when the law was in effect.
- The key treatment is

$$T_{cs} = \text{Law}_s \times \text{CSL}_c,$$

where s indexes state and c indexes birth cohort.

- $\text{Law}_s = 1$ for states with a German ban in the main sample: Indiana and Ohio.
- $\text{CSL}_c = 1$ for birth cohorts young enough to be affected by compulsory schooling when the ban was imposed.

Interpretation: The treatment is cohort-by-state exposure. Older German-Americans in the same law states provide within-state comparisons, and younger German-Americans in no-law states provide cross-state comparisons.

3.2 Linked census border sample

- The paper begins with males born 1880–1916 to German parents and living in the border-county sample in 1920.
- These men are linked forward to the 1930 and 1940 censuses using name, birthplace, and birth-year information.
- The linked sample contains 42,624 unique matched records found in either 1930 or 1940.
- The sample focuses on men because historical census linking is much more reliable for men; women often change surnames after marriage.
- Table 2 summarizes the linked border data set. The 1930-linked sample has marriage and children variables; the 1940-linked sample also has schooling and wage information.

Interpretation: The linked census data allow the paper to observe long-run outcomes decades after the school-language policy. This is central because the outcomes are adult identity choices, not immediate school outcomes.

3.3 German Name Index

- The German Name Index (GNI) measures how distinctively German a first name is.
- The index compares the frequency of a name among German-origin individuals to its frequency among non-German-origin individuals.
- A name used only by German-born individuals receives a high score; a name absent among German-born individuals receives zero.
- The paper uses two naming outcomes: log average GNI of all children and log GNI of the first son.
- Table 3 illustrates the index. Names such as Hans, Karl, Otto, August, Christian, Adolph, Gustave, Emil, Rudolph, and Herman have high GNI scores, while names such as Clyde, Russell, Melvin, Patrick, and Leroy have zero GNI.

Interpretation: First names are useful because they are intentional parental choices. If forced monolingual schooling assimilated parents, their children should receive less German names. If it triggered backlash, their children should receive more German names.

3.4 Inter-marriage

- Inter-marriage is measured as whether the spouse is of German ancestry.
- The paper treats marriage within the German ethnic group as a measure of ethnic attachment.
- Spouse ethnicity is observed in linked census data.

- Intermarriage is an equilibrium outcome: it reflects preferences, local marriage-market composition, and social network constraints.

Interpretation: Endogamous marriage is less directly chosen than a child’s name, but it is a strong behavioural measure of integration. More endogamy after the ban indicates stronger attachment to the ethnic group.

3.5 World War II volunteering

- The paper links men to WWII Army enlistment records.
- Volunteering is identified using Army serial-number ranges. Volunteers had serial numbers in the 11–19 million series.
- The sample is restricted to enlistments between 1940 and 1942, before voluntary enlistment ended by presidential order.
- Table 4 summarizes enlistment data: German-parent men have lower volunteer rates than the broader linked enlistment sample.

Interpretation: Volunteering is a powerful outcome in this historical setting because German-Americans in WWII faced a symbolic choice involving the host country and the ancestral country. Lower volunteering is interpreted as weaker identification with the United States.

3.6 Lutheran Church–Missouri Synod data

- To study mechanisms, the paper uses Statistical Yearbooks of the Lutheran Church–Missouri Synod.
- The data measure church activity in 1916 and 1921 at the county level.
- Outcomes include schools, teachers, pupils, Sunday-school pupils, and German-language monthly services.
- The Lutheran Church–Missouri Synod is useful because it had strong German roots and emphasized German-language parochial schooling.

Interpretation: The church data help distinguish parental demand for ethnic enculturation from a supply-side response by ethnic institutions. A rise in Sunday-school pupils is closer to parental demand than a rise in church infrastructure.

4 Empirical specifications and identification

4.1 Baseline difference-in-differences specification

The baseline specification is

$$Y_{isc} = \alpha + \beta T_{cs} + \lambda_c + \theta_s + \delta Z_{isc} + \varepsilon_{isc}, \quad (1)$$

where Y_{isc} is an identity outcome for individual i in state s and cohort c , $T_{cs} = \text{Law}_s \times \text{CSL}_c$, λ_c are cohort fixed effects, θ_s are state fixed effects, and Z_{isc} includes name-string controls used to improve census-linking comparability.

Interpretation: The coefficient β compares younger treated cohorts in ban states to older cohorts in the same states and to similar cohorts in no-ban neighbouring states.

4.2 Controls for differential trends

A more conservative specification includes interactions between pre-treatment state characteristics and treated-cohort indicators:

$$Y_{isc} = \alpha + \beta T_{cs} + \lambda_c + \theta_s + \delta Z_{isc} + \gamma (\text{GermanShare}_s \times D_c) + \varepsilon_{isc}. \quad (2)$$

- D_c is an indicator for treated cohorts.
- GermanShare_s is the pre-treatment share of Germans in the state.
- Some specifications add county fixed effects and state-specific linear trends.

Interpretation: These controls address the concern that states with stronger German communities may have had different cohort trends even without a language ban.

4.3 Identification assumption and threats

- The identifying assumption is that, absent the ban, treated cohorts in Indiana and Ohio would have followed trends similar to comparable cohorts in neighbouring no-law states.
- Border counties make this assumption more plausible by narrowing the comparison to nearby places.
- Older cohorts provide a pre-treatment comparison within the same law states.
- Pre-trend plots test whether older cohorts in law and no-law states already differed in the outcomes before the affected cohorts appear.
- The paper addresses threats from anti-German discrimination, other state laws, confounding parochial-school policies, selective migration, alternative GNI calculations, and inference with few treated clusters.

Interpretation: The design is not randomized. Its strength comes from combining geography, cohort exposure, fixed effects, controls, falsification tests, and robustness checks.

5 Main results

5.1 Naming and marriage outcomes

- Figure 3 shows the density of log GNI for the first son. In treated cohorts, the law-state distribution is shifted toward more German names.
- Table 5 estimates the baseline border specification.
- For log average GNI of children, the preferred estimates are positive and statistically significant, around 0.79–0.92 depending on controls.
- For log GNI of first son, the effect is larger, around 1.31–1.64 depending on controls.
- For spouse German, the effect is around 3.6–5.7 percentage points, indicating higher ethnic endogamy.

Interpretation: These results directly contradict the assimilation interpretation. If the German ban worked as intended, names should become less German and intermarriage should increase. Instead, names become more German and endogamy rises.

5.2 WWII volunteering

- Figure 4 shows that younger no-law cohorts have higher volunteering rates than younger law cohorts.
- Table 6 estimates a negative treatment effect on volunteering: about -2.4 to -3.7 percentage points in the baseline German-parent sample.
- The effect is large relative to the average volunteering rate of roughly 11% among younger cohorts.
- As a placebo, Italian-origin men do not show the same negative effect.

Interpretation: Volunteering reinforces the identity interpretation. A school policy meant to Americanize German-Americans appears to have reduced their willingness to voluntarily serve the United States in WWII.

5.3 Pre-trends and robustness

- Figure 5 plots treatment effects by birth-cohort bins for all main outcomes.
- Older cohorts show no strong differential trends before the affected cohorts enter school age.
- Robustness checks use alternative control groups, alternative GNI construction, controls for other laws, checks for selective migration, and inference corrections.
- The main pattern remains: naming and marriage results move in the backlash direction; volunteering falls.

Interpretation: The robustness evidence increases credibility by showing that the results are not driven by pre-existing cohort trends or one particular measurement choice.

6 Mechanisms

6.1 Cultural-transmission model intuition

- The paper uses a model inspired by Bisin and Verdier’s intergenerational cultural transmission framework.
- Parents care about children sharing their cultural identity.
- Children can acquire ethnic identity through parents, schools, peers, churches, and community networks.
- If school transmission of German identity is removed, parents with strong ethnic preferences may substitute toward home and community investment.
- If this private investment is strong enough, the policy can reverse its intended effect and strengthen minority identity.

Interpretation: The key idea is substitution between schooling and parental socialization. A language ban removes German from school but may raise the marginal value of parental enculturation.

6.2 Parents’ ethnic background

- Table 7 extends the sample to men with a German father and a non-German mother.
- The baseline law effect remains positive for those with German parents, but the interaction with non-German mother is negative for naming and endogamy outcomes.
- The backlash is therefore weaker or reversed for children of mixed couples.

Interpretation: Mixed-parent households likely have weaker German identity or stronger attachment to mainstream culture. The weaker response in those households supports the parental-identity mechanism.

6.3 Strength of local identity

- Figure 6 and Table 8 show that backlash is stronger in counties with higher shares of Lutheran church members.
- The Lutheran Church–Missouri Synod was strongly associated with German language and parochial schooling.
- Figure 6 and Table 8 also show that the backlash is stronger where the German share is smaller.
- This pattern fits the model: when Germans are a small minority, children are less likely to absorb German identity from peers, so parents invest more intensely.

Interpretation: Community identity matters. Backlash is strongest where identity is intense and where parents cannot rely on peer socialization to transmit German culture.

6.4 Parents and churches

- Figure 7 and Table 9 study Lutheran Church–Missouri Synod activity from 1916 to 1921.
- Law states show a larger increase in Sunday-school pupils.
- Other church activities such as schools, teachers, and pupils do not increase in the same way.
- The share of monthly services in German is somewhat lower in law states, not higher.

Interpretation: The church evidence suggests that parents increased demand for German enculturation through Sunday schools, rather than churches simply expanding all German-language activities.

6.5 Education and labour market costs

- Table 10 studies educational attainment and wage income.
- The ban is associated with a small increase in years of schooling, about 0.31 years.
- The estimated wage effect is negative, around -0.106 log points, but not statistically significant.
- The schooling result rules out a simple story in which backlash outcomes are caused by lower school quality.

Interpretation: If anything, the policy slightly increased formal schooling. The identity outcomes therefore likely reflect cultural response, not reduced educational achievement.

7 Tables and figures

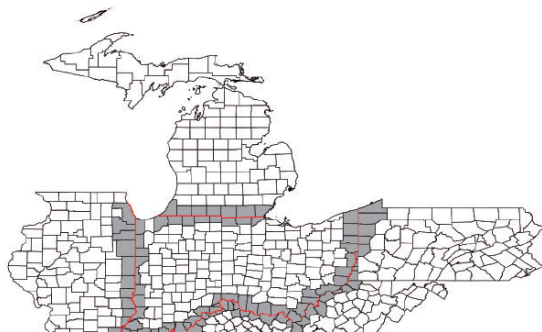
7.1 Figure 1 and Table 1: border design and balance

Read:

- Figure 1 maps the border counties around Indiana and Ohio.
- Table 1 shows that law and no-law border counties are broadly balanced on population density, urban share, foreign-born share, German-born share, and Lutheran share.

Economic message: The geographic design narrows the comparison to similar places and makes the treatment-control contrast more credible.

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(a) Border counties.

TABLE 1
Balancedness of border counties

	No Law	Law	Difference
Population density	144.651 (347.615)	121.4397 (177.746)	23.21 (52.49)
Share urban	0.266 (0.267)	0.330 (0.280)	-0.064 (0.054)
Share foreign-born	0.058 (0.075)	0.071 (0.080)	-0.013 (0.015)
Share German-born	0.018 (0.021)	0.022 (0.024)	-0.005 (0.004)
Share Lutheran	0.012 (0.022)	0.012 (0.019)	0.00003 (0.004)
Observations	58	47	

Notes: Data are from the 1910 county data in [ICPSR](#) and from the 1906 Census of Religious Bodies.

(b) Balancedness.

7.2 Table 2, Figure 2, and Table 3: linked data and naming measure

Read:

- Table 2 reports summary statistics for men linked to 1930 or 1940.
- Figure 2 maps linked individuals in 1920 across law and no-law counties.
- Table 3 shows the GNI interpretation: highly German names have high index values; non-German names have zero values.

Economic message: The identity outcomes are observed behaviours in linked administrative and census data.

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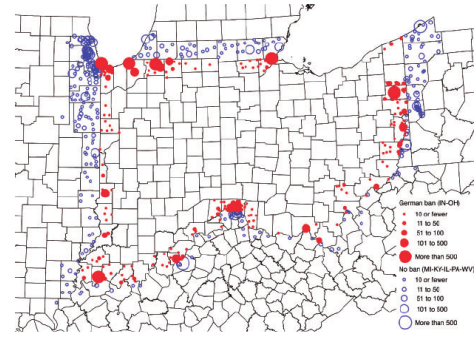
REVIEW OF ECONOMIC STUDIES

TABLE 2
Summary statistics: border data set

	Found in 1930			Found in 1940		
	Mean	S.D.	N	Mean	S.D.	N
Married	0.695	0.460	35,886	0.790	0.407	41,875
Spouse of German ancestry	0.395	0.489	24,925	–	–	–
Number of children	2.398	1.543	20,008	2.206	1.450	25,393
Log average GNI of children	1.774	3.153	19,684	1.305	3.497	20,098
Log GNI of first son	1.358	3.662	14,309	1.095	3.775	12,816
Years of education	–	–	–	8.546	2.595	25,103
Log yearly salary earnings	–	–	–	6.920	2.144	19,305

Notes: The table shows summary statistics for males born 1880–1916 to German parents, who in 1970 lived in a county on

(a) Linked sample.

FIGURE 2
Locations of linked data set in 1920

Notes: The map shows the town-level location of all males, who were born 1880–1916 to German parents, were living in a border county in 1920 and who could be linked to the 1930 or 1940 census.

(b) Linked locations.

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REVIEW OF ECONOMIC STUDIES

TABLE 3
Most and least German-sounding names in the 1930 census

Name	Highest-scoring			Name	Lowest-scoring		
	Total	Germans	GNI		Total	Germans	GNI
Hans	1281	325	98.91	Clyde	7359	0	0
Karl	1542	267	96.21	Russell	6049	0	0
Otto	5693	962	94.78	Melvin	5687	0	0
August	5761	1264	92.30	Patrick	5381	0	0
Christian	1215	179	90.99	Leroy	5187	0	0
Adolph	3231	385	90.27	Warren	5077	0	0
Gustave	1271	231	90.21	Marvin	4588	0	0
Emil	4261	515	89.52	Jim	4231	0	0
Rudolph	3372	296	89.06	Glenn	3898	0	0
Herman	11446	1399	88.27	Leslie	3796	0	0

Notes: The table shows the values of the German name index for the 10 highest-scoring (left panel) and 10 lowest-scoring (right panel) names of males born in 1910 in the 1930 5% DTIC sample. Highest-scoring names are chosen among

German Name Index examples.

7.3 Table 4 and baseline equations: enlistments and DiD design

Read:

- Table 4 summarizes WWII enlistment variables.
- The baseline equations show the difference-in-differences structure: treatment is being in a law state and in an affected school-age cohort.

Economic message: The enlistment data provide a rare behavioural measure of loyalty and host-country identification.

TABLE 4
Summary statistics: WWII enlistments

	All			German parents		
	Mean	S.D.	N	Mean	S.D.	N
Age	19.629	4.863	160,780	22.358	5.768	2,679
Married	0.011	0.107	160,780	0.014	0.120	2,679
With dependents	0.107	0.309	160,780	0.155	0.362	2,679
Volunteer	0.109	0.312	160,780	0.079	0.270	2,679
High school graduate	0.374	0.484	160,780	0.274	0.446	2,679
College graduate	0.064	0.244	160,780	0.045	0.207	2,679

(a) WWII enlistment summary.

of school age and cohorts too old to be at school between states with and without a law. My main specification takes the form:

$$Y_{isc} = \alpha + \beta T_{cs} + \lambda_c + \theta_s + \delta Z_{isc} + \varepsilon_{isc}, \quad (1)$$

T_{cs} is an indicator for individuals living in a state with a law and who were within the age for compulsory schooling (CSL) at the time that law was in place. The terms λ_c and θ_s are cohort and state of residence (in the case of the border county data set) or birth (in the case of the WWII enlistments data set) fixed effects. Z_{isc} is a vector of name string properties that are correlated with the probability of a record being matched in a later census.¹⁴ The coefficient of interest is δ , the estimated average effect of legislation on exposed cohorts.

In the above specification, state and cohort fixed effects account for average differences in the outcome variable across states and cohorts. As with every DiD approach, the identifying assumption is that there exists no omitted time-varying and state-specific factor correlated with the passage of language laws and with the outcome variables. Because it is difficult to fully rule out this concern in an observational setting, I will report specifications that control for interactions of state-level variables recorded before the enactment of the law and that are likely correlated with its passage (most notably the share of the German element in a state's population) with an indicator for the treated cohort (D_c).

$$Y_{isc} = \alpha + \beta T_{cs} + \lambda_c + \theta_s + \delta Z_{isc} + \gamma \times \text{German share}_s \times D_c + \varepsilon_{isc} \quad (2)$$

These include the length and commonness of the first and last name. Commonness is computed as the share of the 1920 census with the same first or last name.

(b) Baseline DiD equations.

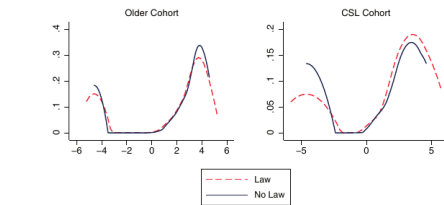
7.4 Figure 3 and Table 5: naming and intermarriage backlash

Read:

- Figure 3 shows that treated cohorts in law states have a higher density of German-sounding first-son names.
- Table 5 shows positive treatment effects on log average GNI, log GNI of first son, and spouse German.

Economic message: This is the core backlash result: a policy intended to weaken German identity instead strengthens German naming and marriage patterns.

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FIGURE 3
Densities of log GNI of first son by cohort

Notes: The figure illustrates, for the linked border data set, the kernel density of the logarithm of the GNI of the first son. The panel plots this density for the cohort too old to have been in school (for compulsory law) at the time German was banned; the right panel plots the density for the treated cohort.

(a) Name density.

TABLE 5 Baseline results: border data set				
	[1]	[2]	[3]	[4]
Panel A: Dep. variable is Log average GNI of children				
Law × CSL age	0.473 (0.279) [0.1191]	0.884** (0.309) [0.0801]	0.920** (0.308) [0.0631]	0.794** (0.334) [0.1291]
Observations	26334	26334	26334	26334
R-squared	0.0494	0.0515	0.0547	0.0553
Panel B: Dep. variable is Log GNI of first son				
Law × CSL age	1.231*** (0.280) [0.00200]	1.615*** (0.300) [0.0230]	1.638*** (0.299) [0.0230]	1.310*** (0.241) [0.0240]
Observations	18459	18459	18459	18459
R-squared	0.0162	0.0194	0.0290	0.0295
Panel C: Dep. variable is Spouse German				
Law × CSL age	0.0472** (0.0164) [0.0781]	0.0573*** (0.0183) [0.0691]	0.0365** (0.0148) [0.0941]	0.0389*** (0.0105) [0.0290]
Observations	24925	24921	24921	24921
R-squared	0.0510	0.0523	0.0634	0.0636
Additional controls	N	Y	Y	Y
County FE	N	N	Y	Y
State trends	N	N	N	Y

Notes: The sample consists of males, born 1880–1916 in the U.S. to German parents, living in a border county in 1920 and who were linked to the 1930 census (Panel C) or the 1930 and 1940 census (Panels A and B). All regressions include residence state in 1920 and birth cohort fixed effects, and controls for the following name string properties: first and last name length and first and last name commonness. Regressions in Panels A and B include a census year indicator. Additional controls in Panels A and B include the share of Germans in the state in 1910 and border segment indicators, all interacted with an indicator for the treated cohort. In Panel C, they include the number of first and second generation German women born in the 10 years before (and including) each birth cohort. Column [4] controls for a linear state-specific trend fitted to the control cohorts. Standard errors are clustered at the state × border segment level. p -values from the wild bootstrap (Cameron *et al.*, 2008) are reported in brackets. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

(b) Baseline results.

7.5 Figure 4 and Table 6: WWII volunteering

Read:

- Figure 4 shows volunteering rates by cohort and law status.
- Table 6 estimates a negative treatment effect on volunteering for Germans.

Economic message: The volunteering evidence shows that backlash appears in a public act of host-country identification.

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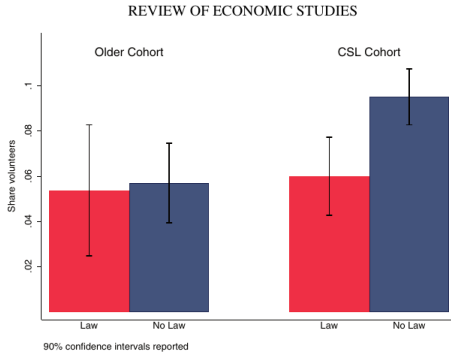


FIGURE 4
Share of volunteers by cohort and law status
Notes: The bars on the left show the share of U.S. Army volunteers by language law status for the cohort too old to have been in (by compulsory law) at the time German was banned; the bars on the right plot the respective share for the treated cohort.

(a) Volunteer shares.

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TABLE 6 Baseline results: WWII enlistments						
Dep. variable:	[1]	[2]	[3]	[4]	[5]	[6]
Volunteer						
Law × CSL age	−0.0256** (0.00696) [0.004]	−0.0235** (0.00928) [0.004]	−0.0243** (0.00986) [0.004]	−0.0370*** (0.00785) [0.460]	0.00595 (0.0414) [0.903]	−0.00389 (0.00784) [0.768]
Law × CSL age × German parents						−0.0313* (0.0149) [0.116]
Observations	2679	2667	2667	2667	5443	160246
R-squared	0.0261	0.0641	0.0643	0.0698	0.0777	0.0746
Additional controls	N	Y	Y	Y	Y	Y
Share German in state in 1910 × Cohort FE	N	N	Y	Y	N	N
State trends	N	N	N	Y	N	N

Notes: The sample consists of males born 1880–1916 in Indiana, Ohio, Illinois, Michigan, Kentucky, West Virginia, or Pennsylvania, who enlisted in the U.S. Army between 1940 and 1942 and who were linked to the 1930 census. In columns [1]–[4], it is restricted to individuals with German parents and in column [5] to individuals with Italian parents. All regressions include state-of-birth and birth cohort fixed effects and control for the following name string properties: first and last name length and first and last name commonness. Column [3] includes an interaction of the share of Germans in the state in 1910 with an indicator for the treated cohort. Columns [2]–[6] control for marital status, the number of dependent family members, and enlistment year fixed effects. Column [4] controls for a linear state-specific trend fitted to the control cohorts. Standard errors are clustered at the state level. *p*-values from the wild bootstrap (Cameron *et al.*,

(b) WWII enlistment regressions.

7.6 Figure 5: pre-trends and cohort-bin estimates

Read:

- Figure 5 plots cohort-bin treatment effects for naming, intermarriage, and volunteering.
- Older cohorts generally do not show large pre-existing differences.

Economic message: The cohort-bin evidence supports the difference-in-differences interpretation.

7.7 Table 7: mixed ancestry and parental identity

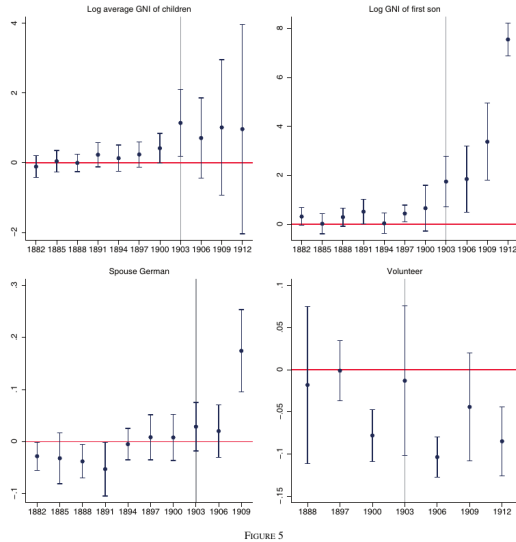
Read:

- Table 7 adds individuals with a German father and non-German mother.
- The backlash effect is weaker or reversed for mixed-parent households.

Economic message: The result links the policy response to parental identity.

7.8 Figure 6 and Table 8: community identity strength

Read:

FIGURE 5
Estimated effects of language ban by birth cohort bin

Notes: The figure shows coefficient estimates and 95% confidence intervals from a regression of each outcome on state and birth cohort fixed effects and a set of interactions of 3-year birth cohort bins with an indicator for a language ban. The vertical line in year 1903 indicates the first cohort to be affected by the language ban.

4.3. Robustness

- Figure 6 shows that treatment effects rise with Lutheran share and fall with German population share.
- Table 8 confirms this pattern in regressions.

Economic message: Community identity matters. Backlash is strongest where identity is intense and where peer socialization is limited.

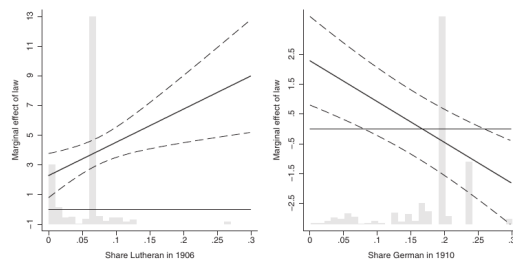


FIGURE 6

Ethnic and religious composition and effects of the language ban

Notes: The left panel of the figure plots coefficient β_2 from a regression specified in equation (4) against the county-level share of Lutheran church members in 1906. The right panel plots coefficient β_3 of the same equation against the share of first- and second-generation Germans in a county in 1910. The dependent variable is the logarithm of the GNI of the first son. Dashed lines represent 95% confidence intervals. The underlaid histograms show how the data are distributed across counties with different shares of Lutheran church members.

(a) Effects by shares.

TABLE 8
Effects by share of Lutheran church members and share of Germans in the county

Dep. variable:	[1] Log average GNI of children	[2] Log GNI of first son	[3] Spouse German
Law $\times$ CSL age	1.991*** (0.629) [0.000]	1.481** (0.575) [0.0840]	0.0220 (0.0497) [0.664]
Law $\times$ CSL age $\times$ Share German	-13.84*** (0.004)	-8.931** (0.0480)	-0.0801 (0.0684)
Law $\times$ CSL age $\times$ Share Lutheran	27.85** (0.0480)	13.73* (0.144)	0.772* (0.108)
Observations	18,459	26,334	24,921
R-squared	0.0248	0.0552	0.0609

Notes: The sample consists of linked males, born 1880–1916 in the U.S. to a German father, living in a border county in 1920 (columns [1]–[3]) or born in any of the seven states in the sample (column [4]). All regressions include (1920 residence or birth) state and birth cohort fixed effects, county fixed effects, and controls for the following name string properties: first and last name length and first and last name commonness. *Share German* and *Share Lutheran* are the share of first- and second-generation Germans and of Lutheran church members in the county in 1910 and 1906, respectively. The latter variable is from the 1906 Census of Religious Bodies. Standard errors are clustered at the state $\times$ border segment level. p -values from the wild bootstrap (Cameron *et al.*, 2008) are reported in brackets. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

(b) Heterogeneity regressions.

7.9 Figure 7 and Table 9: Lutheran church activity

Read:

- Figure 7 shows changes in Lutheran Church–Missouri Synod activity between 1916 and 1921.

- Table 9 estimates corresponding difference-in-differences regressions.

Economic message: The church evidence points to parental substitution into religious/ethnic socialization.

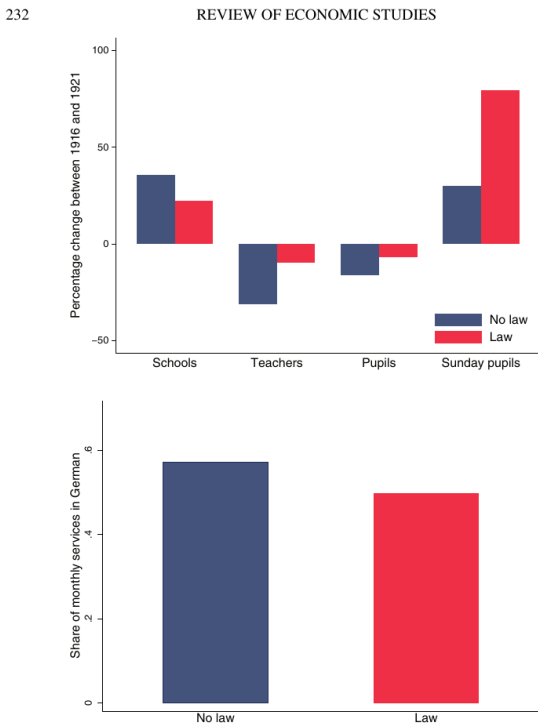


FIGURE 7
Change in the activities of the Lutheran Church-Missouri Synod, 1916–1921
Notes: Data are from the 1916 and 1921 Statistical Yearbooks of the Lutheran Church-Missouri Synod, for the seven states in the s (Indiana, Ohio, Illinois, Michigan, Kentucky, West Virginia, and Pennsylvania).

their language in public school, a German father responds: “German ei, you bet, /This he son! hears from my mouth./For that we don’t need anv teachers.”²⁶

(a) LCMS activity changes.

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TABLE 9
Change in activities of Lutheran Church-Missouri Synod, 1916–1921

	[1]	[2]	[3]	[4]
Dep. variable	Schools	Teachers	Pupils	Sunday pupils
Law × 1921	0.0955 (0.125) [0.3624]	0.155 (0.145) [0.1892]	0.131 (0.123) [0.3674]	0.307 (0.209) [0.0971]
Observations	78	78	78	78
R-squared	0.861	0.873	0.912	0.783

(b) LCMS regressions.

7.10 Table 10: education and earnings

Read:

- Table 10 shows that the ban increased years of schooling by about 0.31 years.
- The estimated wage effect is negative but not statistically significant.

Economic message: The backlash appears cultural rather than educational.

TABLE 10
Educational and labour market outcomes

Dep. variable	[1]	[2]
	Years of schooling	Log yearly wage income
Law $\times$ CSL age	0.311** (0.129) [0.2372]	-0.106 (0.132) [0.6086]
Observations	25,819	19,727
R-squared	0.0678	0.0197

Notes: The sample consists of males, born 1880–1916 in the U.S. to German parents, living in a border county in 1920 and who were linked to the 1940 census. All regressions include residence state in 1920 and birth cohort fixed effects, county fixed effects, indicators for the share of Germans in the state in 1910, and border segment indicators interacted with an indicator for the treated cohort, and controls for the following name string properties: first and last name length and first and last name commonness. When the dependent variable is log yearly wage income, the data set is restricted

Educational and labour-market outcomes.

8 Short summary

- The paper studies German language bans in U.S. schools after World War I.
- It asks whether forced assimilation policies increase integration or trigger backlash.
- The design compares affected school-age cohorts to older cohorts, across law and no-law border states.
- The main data link 1920 German-American men to 1930/1940 census records and WWII enlistment data.
- German identity is measured through children's names, spouse ethnicity, and WWII volunteering.
- Treated cohorts give children more German names and marry Germans more often.
- Treated cohorts are also less likely to volunteer for WWII service.
- The effects are weaker for mixed-parent households.
- The effects are stronger in counties with more Lutheran church members.
- The effects are stronger where Germans are a smaller share of the local population.
- Lutheran Sunday-school enrolment rises after the ban in law states.
- Schooling does not fall; it rises slightly.
- Wages weakly decline, suggesting possible costs of intensified ethnic separation.
- The main conclusion is that forced assimilation can backfire by motivating families to strengthen cultural transmission.

9 Conclusion

9.1 Core conclusion

Fouka shows that prohibiting German in schools after World War I did not assimilate German-American children. Instead, it led to a backlash in cultural identity. The children of German immigrants exposed to the ban later selected more German names for their own children, married within the German ethnic group more often, and volunteered for the U.S. Army at lower rates during World War II.

9.2 Economic intuition

The economic logic is a substitution mechanism in cultural transmission. Schools previously helped transmit German identity. When the state removed that channel, parents with strong ethnic preferences responded by investing more in private or community-based socialization. If this parental response is strong enough, the policy's effect reverses: a policy designed to weaken minority identity strengthens it.

9.3 Takeaway

The paper's broader lesson is that assimilation policies are not mechanical inputs into national integration. Their effects depend on how minority families interpret and respond to them. Coercive policies can make cultural identity more salient, intensify parental investment, and deepen separation from the mainstream. For policy design, the key implication is that integration through repression can be counterproductive.